

Representational Measurement of Similarity: The Additive Difference Model, Revisited

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In dieser Ausarbeitung geht es darum, wie man eine Metrik auf ordinalen Ähnlichkeitsdaten herleiten kann, die zusätzlich additiv ist und durch das additive difference model repräsentiert werden kann. Das Hauptresultat ist, dass die l^p Metrik die einzige additive-difference Metrik, die additiv auf Liniensegmenten ist, was eng mit der Homogenität dieser Metrik zusammenhängt. Zum Herleiten dieses Hauptresultates werden die Grundelemente von Messtheorie, sowie von proximity measurement mithilfe von geometrische Modellen eingeführt. Diese Ausarbeitung basiert auf dem Paper Tversky and Krantz, 1970 und den Büchern Suppes, 2014 und Breinbauer, 1976.

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2 Proximity Structures and Metrics

3 Additive Difference Models and Structures

3.1 Additive Difference Model

3.2 Additive Difference Structure

3.3 Examples

4 Segmental Additivity

4.1 Segmentally Additive Metrics

4.2 Segmentally Additive Proximity Structures

4.3 Examples

5 Combining Segmental Additivity with the AD-Model

Theorem 5.1 (Necessary and sufficient conditions for additive difference metrics).

Suppose $\langle A, \succsim \rangle$ is an additive-difference proximity structure such that f is the additive-difference representation from ??, i.e.,

$$f(a, b) = \sum_{i=1}^n F_i(|\varphi_i(a) - \varphi_i(b)|).$$

Suppose further that the domain of each F_i is a real interval and $F_i(0) = 0$, $i = 1, \dots, n$ (through transformation, since F_i are interval scales).

For $\langle A, \succsim \rangle$ to be compatible with a metric, and additionally be additive on the coordinate axes,

1. it is necessary that there exists a continuous, convex function F , such that $F_i(x) = F(t_i x)$, $t_i > 0$, for all x in the domain of F_i , $i = 1, \dots, n$; and
2. sufficient that, in addition (to the conditions in 1.), the function $\log F(e^x)$ is convex.

Remark 5.2 (Additivity on the coordinate axes). Additivity on the coordinate axes means the following: If $a, b, c \in A$ differ only on one factor A_i , then the triangle inequality of the metric holds with equality. In mathematical terms:

$$a_j = b_j = c_j \text{ for all } j \neq i \text{ and } (a, c) \succsim (a, b), (b, c) \Rightarrow \delta(a, b) + \delta(b, c) = \delta(a, c)$$

In the following lemmas until the proof of this theorem is finished everything will be assumed as in theorem 5.1. **Todo: checken bis wohin genau**

The following remark ties this result closer to the Minkowski metric, and provides a form which the metric must take. This will play an important role in the main theorem.

Remark 5.3 (Form of δ). Examining section 5 more closely yields, that $G = F^{-1}$ must hold and the representation of δ takes the form

$$\delta(a, b) = F^{-1} \left\{ \sum_{i=1}^n F(|\varphi_i(a_i) - \varphi_i(b_i)|) \right\}$$

where F is continuous, convex and increasing.

This follows from two arguments. Firstly, incorporating the t_i into the φ_i

$$\begin{aligned} G \left\{ \sum_{i=1}^n F_i(|\varphi_i(a_i) - \varphi_i(b_i)|) \right\} &= G \left\{ \sum_{i=1}^n F(t_i |\varphi_i(a_i) - \varphi_i(b_i)|) \right\} \\ &= G \left\{ \sum_{i=1}^n F(|\varphi_i(a_i) - \varphi_i(b_i)|) \right\} \end{aligned}$$

yields the desired form inside the sum.

Secondly, by using $F_i(x) = F_n(t_i x)$ and $F_i(x) = G^{-1}(t_i x)$. The first equation applied for $i = n$ yields $F_n(x) = F_n(t_n x)$ which implies $t_n = 1$. Using this in the second equation for $i = n$ yields $F(x) = F_n(x) = G^{-1}(t_n x) = G^{-1}(x)$. Hence, by taking the inverse on both sides, we get the desired form for δ .

Proof of theorem 5.1 1. According to the assumptions of the theorem, the additive difference representation $f(a, b)$ is compatible with a metric δ on A , i.e.,

$$f(a, b) \leq f(c, d) \iff \delta(a, b) \leq \delta(c, d) \quad \text{for all } a, b, c, d \in A.$$

Therefore, there exists an increasing function G , such that for all $a, b \in A$

$$\delta(a, b) = G \left(\sum_{i=1}^n F_i (|\varphi_i(a_i) - \varphi_i(b_i)|) \right).$$

Another possible perspective on this is, that by **Todo: ref** f is an interval scale, therefore also is an ordinal scale. Since the ordinal ordering of all pairs of elements from A is same for f and δ , there must be an increasing function which transforms one scale into the other and preserves the ordinal orderings. Assume a , b , and c differ only in the i -th factor and $a|b|c$. Thus, we can use additivity on the i -th coordinate axis for a, b, c which yields

$$G(F_i(|\varphi_i(a_i) - \varphi_i(b_i)|)) + G(F_i(|\varphi_i(b_i) - \varphi_i(c_i)|)) = G(F_i(|\varphi_i(a_i) - \varphi_i(c_i)|)).$$

Defining $|\varphi_i(a_i) - \varphi_i(b_i)| =: x$ and $|\varphi_i(b_i) - \varphi_i(c_i)| =: y$ yields

$$G[F_i(x)] + G[F_i(y)] = G[F_i(x + y)].$$

Further, define $H_i(x) = G[F_i(x)]$. This reduces the above equation to $H_i(x) + H_i(y) = H_i(x + y)$ for all x, y in the domain of F_i , $1 \leq i \leq n$. This means, that H_i must be linear and since the values are positive, the only monotonic solution to this equation is $H_i(x) = t_i x$, $t_i > 0$. Since G is increasing, it has an inverse G^{-1} satisfying

$$G^{-1}[H_i(x)] = G^{-1}[t_i x] = F_i(x)$$

where F_i is defined. Since G^{-1} is independent of i , the F_i can only differ in range and in the unit of the domain (i.e. the t_i which adjusts the domain where G^{-1} is defined). Without loss of generality, we can assume that the range of F_n is the largest. Since the range of F_n includes the range of F_i , we can set $F = F_n$ and obtain $F_i(x) = F_n(t_i x) = F(t_i x)$ for $t_i > 0$ and for every x in the domain of F_i . Continuity and convexity will be proven in the next lemmas. $\square$

Todo: sagen dass conditions wie oben sind

Lemma 5.4.

$$F(\alpha + \beta + \gamma) + F(\gamma) \geq F(\alpha + \gamma) + F(\beta + \gamma).$$

Proof. In the triangle inequality,

$$F^{-1} \left[\sum_{i=1}^n F(\alpha_i) \right] + F^{-1} \left[\sum_{i=1}^n F(\beta_i) \right] \geq F^{-1} \left[\sum_{i=1}^n F(\alpha_i + \beta_i) \right],$$

let $\alpha_1 = \gamma, \alpha_2 = F^{-1}(F(\alpha + \gamma) - F(\gamma)), \alpha_j = 0$ for $3 \leq j \leq n$ and $\beta_1 = \beta, \beta_j = 0$ for $2 \leq j \leq n$. Hence,

$$\begin{aligned} F(\alpha + \beta + \gamma) &= F \left[F^{-1} \left[\sum_{i=1}^n F(\alpha_i) \right] + F^{-1} \left[\sum_{i=1}^n F(\beta_i) \right] \right] \\ &\geq \sum_{i=1}^n F(\alpha_i + \beta_i) \quad (\text{by the triangle inequality}) \\ &= F(\beta + \gamma) + F(\alpha + \gamma) - F(\gamma). \end{aligned}$$

□

Lemma 5.5.

F is continuous.

Proof. Since F is increasing, all discontinuities are gaps. Real analysis yields, that real monotonic functions have at most countable many gaps. We will derive a contradiction from this.

Assume $F(\alpha)$ is the lower boundary of a gap, then there exists $\epsilon > 0$ such that for all $\delta > 0$, $F(\alpha + \delta) - F(\alpha) > \epsilon$. However, by lemma 5.4, for every $\beta > 0$,

$$F(\alpha + \beta + \delta) - F(\alpha + \beta) \geq F(\alpha + \delta) - F(\alpha) > \epsilon,$$

and so there is a gap at any $\alpha + \beta$, and F would have to be discontinuous everywhere. This is impossible, so F must be continuous. □

Lemma 5.6 (Characterization of Convexity).

Suppose F is a strictly increasing function defined on the nonnegative reals. Then F is convex iff for all $\alpha, \beta, \gamma \geq 0$ the inequality from 5.4 holds, i.e.,

$$F(\alpha + \beta + \gamma) + F(\gamma) \geq F(\alpha + \gamma) + F(\beta + \gamma).$$

Proof.

Part 1 ($\Rightarrow$).

Suppose, F is convex, and let $p = \alpha/(\alpha + \beta)$, then

$$\begin{aligned} p(\alpha + \beta + \gamma) + (1 - p)\gamma &= \alpha + \gamma \\ (1 - p)(\alpha + \beta + \gamma) + p\gamma &= \beta + \gamma. \end{aligned}$$

Hence, by convexity,

$$\begin{aligned} F(\alpha + \gamma) + F(\beta + \gamma) &= F[p(\alpha + \beta + \gamma) + (1 - p)\gamma] + F[(1 - p)(\alpha + \beta + \gamma) + p\gamma] \\ &\leq pF(\alpha + \beta + \gamma) + (1 - p)F(\gamma) + (1 - p)F(\alpha + \beta + \gamma) + pF(\gamma) \\ &= F(\alpha + \beta + \gamma) + F(\gamma). \end{aligned}$$

Part 2 ($\Leftarrow$).

To prove the converse, we will use the continuity of F by lemma 5.5. Suppose, $\alpha > \beta$, then

$$\begin{aligned} F(\alpha) + F(\beta) &= F\left(\frac{\alpha - \beta}{2} + \frac{\alpha - \beta}{2} + \beta\right) + F(\beta) \\ &\geq 2F\left(\frac{\alpha - \beta}{2} + \beta\right) && \text{(by lemma 5.5)} \\ &= 2F\left(\frac{\alpha + \beta}{2}\right), \end{aligned}$$

which is equivalent to convexity for a continuous F .

□

5.1.1 Convexity of δ Balls

6 Discussion

References

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